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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: PRESSURE RELIEF VALVES (API 520 / API 526)

Document Code: DS-PS-0018 | Revision: Z1 | Status: Issued for Construction

Process Unit: Unit 2300 (Decomposition & Recovery Section)

1. PSV-2304A / PSV-2304B (PROTECTING DECOMPOSER REACTOR D-2304)

- Protected Equipment: D-2304 (Decomposer Reactor)
- Tag Numbers: PSV-2304A (Active) / PSV-2304B (100% Spare via 3-Way Selector Valve)
- Valve Type: Balanced Bellows Spring-Loaded Relief Valve with Rupture Pin Buckling Assist
- Set Pressure: 11.0 kg/cm2g (Matches DS-D2304 Design Pressure of 11.0 kg/cm2g; Note: P&ID PID-23-0013 note states)
- Relieving Temperature: 185.0 degC (Two-Phase Flashing Phenol/Acetone/Cumene Mixture)
- Governing Sizing Case: Exothermic Runaway / Cooling Loss on E-2307 (DIERS Two-Phase Method per STD-ENG-014)
- Required Relieving Rate: 38,400 kg/hr
- Selected API 526 Orifice: 4J6 (4 in 300# Inlet x 6 in 150# Outlet, Effective Area 8.303 cm2)
- Body & Trim Material: 316L Stainless Steel / Hastelloy C-276 Bellows
- Discharge Destination: Acid Aromatics Relief Header to Knock-Out Drum D-2320 (PID-23-0022)

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DS-PS-0018 PAGE 2 OF 3 - PSV-2301 (PROTECTING PREFLASH COLUMN V-2301)

- Protected Equipment: V-2301 (Preflash Column)
- Tag Number: PSV-2301
- Valve Type: Balanced Bellows Pressure Relief Valve
- Set Pressure: 3.5 kg/cm2g
- Relieving Temperature: 148.0 degC
- Governing Sizing Case: Blocked Overhead Vapor Outlet (Stream S220) with Reboiler Steam Pinch
- Required Relieving Rate: 19,200 kg/hr
- Selected API 526 Orifice: 3K4 (3 in 150# Inlet x 4 in 150# Outlet)
- Discharge Destination: Closed Relief Header to Knock-Out Drum D-2320 (PID-23-0022)

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DS-PS-0018 PAGE 3 OF 3 - BACKPRESSURE & RUPTURE DISK ISOLATION NOTES

- Maximum Built-Up Backpressure in Relief Header to D-2320: 1.15 kg/cm²g (10.5% of PSV-2304A set pressure, compli
- Upstream Rupture Disk PSE-2304A/B installed beneath PSV-2304A/B to prevent polymer fouling of valve seats.